

Demonstrating Swarm Robots Capable of Cooperative Transitioning between Table and Wall

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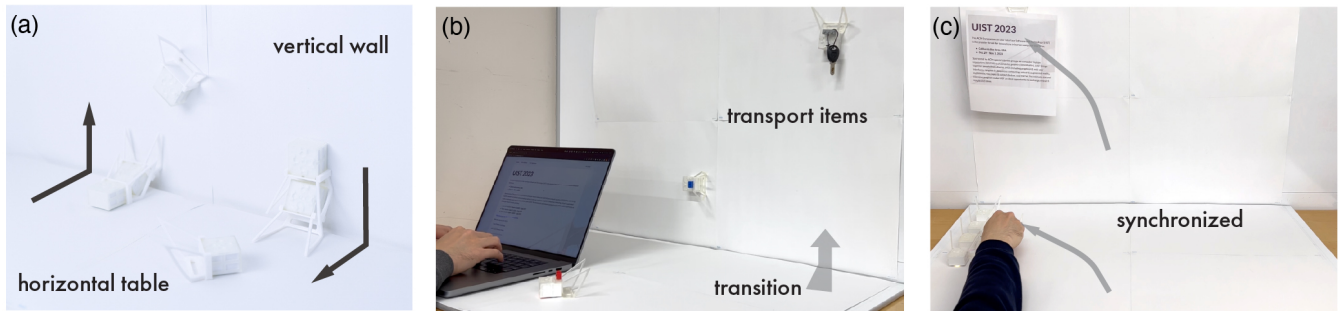


Figure 1: Overview of the system. (a) The robots can cooperatively transit between horizontal and vertical surfaces. Users can use them to (b) transport small items organize workspace, (c) or dynamically arrange wall environment.

ABSTRACT

Swarm User Interfaces (Swarm UIs) enable manipulation of user environment through the dynamic arrangement of small robots. However, the operation range of Swarm UIs are limited to a single plane due to their locomotion constraints, which are typically two-wheel-propelled. Here, We present a proof-of-concept design for swarm robots, which enables them to cooperatively transition between horizontal and vertical surfaces. Notably, this design requires only passive mechanical structure and does not rely on any powered electrical components. We demonstrate several application examples to showcase the feasibility of the robots.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices; Haptic devices.**

KEYWORDS

swarm user interfaces, human robot interaction

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1 INTRODUCTION

Swarm User Interfaces (Swarm UIs), which employ multiple robots as interfaces, have garnered considerable interest, as exemplified by several works [5, 9, 11]. A robot swarm consists of a collection of small robots, each with a simple function, that collaborate together like an animal swarm to achieve various applications. Swarm UIs interact with humans on an individual basis and convey information through the shape of the entire group, with most applications relying on the robots' movement. Thus, enhancing the locomotion capabilities of swarm robots is crucial.

However, although the movement range of swarm robots encompasses wall surfaces [3], ceilings [6, 13] and clothing[1], their primary design is intended for utilization on horizontal surfaces, particularly tabletops [4, 5, 7, 8, 10]. Conversely, while wall surfaces are deemed promising for human-robot interactions [12], to our knowledge, swarm robots capable of transitioning between tabletop and wall surfaces have not yet been developed.

The realization of swarm robots capable of transitioning between tabletop and wall surfaces would extend their operational range from flat areas to three-dimensional spaces that include wall surfaces, enabling multiple robots to cooperatively manipulate the environment. For example, these robots can organize items on the tabletop and relocate them to the wall for storage. One feasible approach for small two-wheeled robots to move between the tabletop and the wall involves installing a ferromagnetic slope between the two surfaces, allowing the robots to adhere during transition. However, this slope might take up substantial desktop space and potentially disrupt with user activities. Ideally, robots should transition using their inherent capabilities, avoiding major alterations to the wall or tabletop environments. Instead, we adopt a strategy wherein multiple robots collaborate to ensure a smooth transition.

In this paper, we present a proof-of-concept design for swarm robots capable of transitioning between tabletop and wall. We devised a 3D-printable attachment that can be effortlessly fabricated and affixed to commercially available two-wheeled robots. It requires only a passive 3D-printed structure, without the need for additional powered electrical components such as motors. Owing to the attachment design, a robot can transition between the two perpendicular surfaces with the assistance of another pushing robot. This approach ensures that transitions can be accomplished utilizing the robots' inherent capabilities, minimizing alterations to wall surfaces or desktop environment.

2 TABLE-WALL TRANSITIONING SWARM ROBOTS

2.1 Transition Mechanism

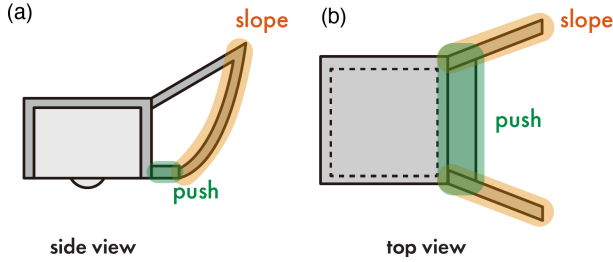


Figure 2: Basic shape of the attachment. It has two key components called *slope* and *push*

We employ two cube-shaped robots with attachments to achieve a 90-degree rotation by pressing the transitioning robot against the wall or table. A preliminary design of the attachment is depicted in Figure 2. The attachment comprises of two key components: *push* and *slope*. The push component pushes the lower rear of the transitioning robot, while the slope component enables a 90-degree rotation when pushed. The push component is located at the front of the robot and between the slope component.

For the transition from the tabletop to the wall, the movement occurs as illustrated in Figure 3 (a). In the following explanation, the assisting robot (blue) is labeled as A, and the transitioning robot (red) as B.

- (1) Align A and B in a straight line on the tabletop close to the transition wall.
- (2) A pushes B, causing B to press against the wall and rotate 90 degrees with the *slope* component.
- (3) B adheres to the ferromagnetic wall with the bottom-attached magnet.
- (4) As the transition is complete, B can maneuver on the wall.

Similar to the table-to-wall case, the wall-to-table transition can happen in reverse. The process is shown in Figure 3 (b).

As all robots have the same universal attachment, they can transition between tables and walls. This could enhance the adaptability of the system, letting one robot substitute for another if a robot malfunctions.

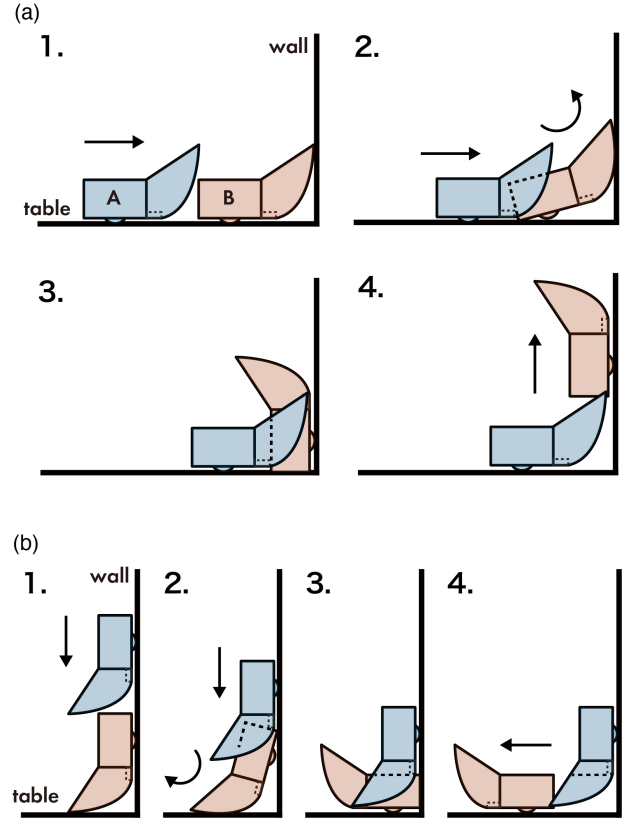


Figure 3: Transition process: (a) table-to-wall transition, (b) wall-to-table transition

2.2 Implementation

We have developed a robot prototype using the commercially available toy platform, *toio*TM [2], displayed in Figure 4. The dimensions of the attachment were designed to avoid the interference between the transitioning and assisting robots. *toio* robots feature LEGO-compatible adapters, so we designed our attachment for an easy fit by simply pressing it onto the robot. Like prior works [3, 7], we added a disc-shaped neodymium magnet to the robot's base to enable wall adherence and increase wheel grip against the tabletop (Figure 4(a)). The magnet is 12 mm in diameter, 1.3 mm thick, and has a surface magnetic flux density of 0.20 T. We added LEGO-compatible adapters on top for payloads or additional functions. As shown in Figure 4(c), extra payload can be added in the green-filled area, free from interference by the assisting robot during transition. We crafted the attachments using an SLA 3D printer (Formlabs Form 3B+ with Clear V4 resin).

3 APPLICATIONS

We showcase the versatility of the system with various application scenarios. These examples emphasize the unique features, including its ability to transition between wall and tabletop surfaces, and facilitate user interaction.

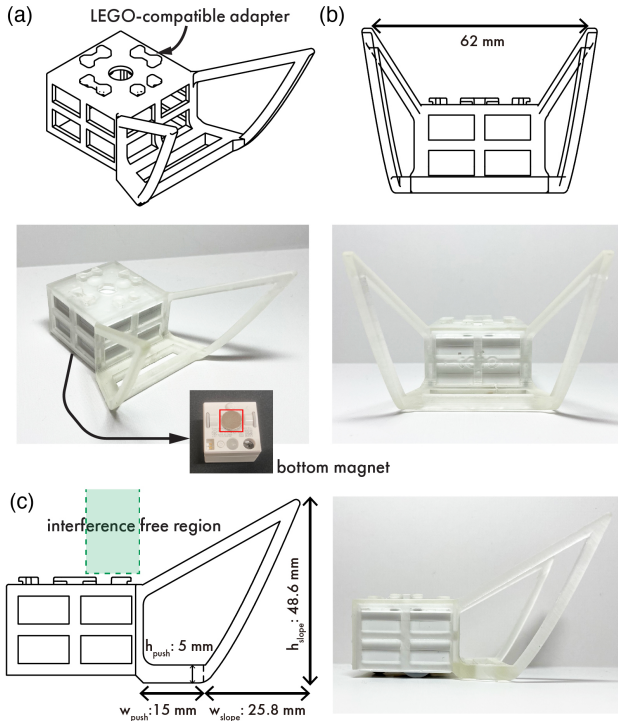


Figure 4: CAD and photos of the fabricated attachment. (a) top-right view, (b) front view, and (c) side view.

3.1 Workspace Organization

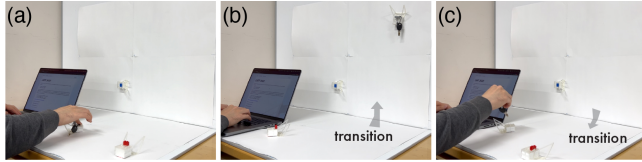


Figure 5: Robots help the user organizing the workspace.

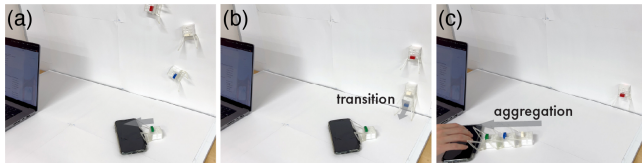


Figure 6: Robots can transition between surfaces to generate an aggregate force for moving heavy objects

The robots can aid in efficient workspace organization, particularly key storage. In this case, the user attaches a key to a table-bound robot (Figure 5(a)). The robot transitions from the table to the wall, storing the key at a designated wall spot (Figure 5(b)). When the key is needed, the robot descends and delivers it to the user (Figure 5(c)).

Normally, robots stay on the wall for better desk space utilization, not hindering the user's workspace. When a table robot tries to deliver a heavy phone to the user and fails (Figure 6(a)), wall-bound robots can descend and help carry the phone (Figure 6(b)(c)).

3.2 Dynamic Manipulation of Wall Objects



Figure 7: Dynamic manipulation of wall objects using the transportation and synchronization feature.

We demonstrate the ability of the robots that dynamically manipulate wall environment in an application with a poster. The user affixes the poster to a robot (Figure 7(a)), which cooperates with another robot to transport the poster to the wall (Figure 7(b)). Then, the table robot acts as a remote control for the wall robot, synchronizing their movements. The user can adjust the poster's position by maneuvering the table robot (Figure 7(c)). This allows the user to effortlessly interact with wall spaces that are typically out of reach.

4 CONCLUSION

We presented a unique robotic system designed for seamless transitions between tabletop and wall surfaces, broadening the potential of mobile robots within Swarm UIs. We also demonstrated a range of applications to showcase the feasibility of the system in workspace organization and wall environment. Moving forward, we plan to explore the potential of the system in widening spatial opportunities and enhancing human-robot interaction.

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